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Batch: A-2
Division: A

Subject: Programming with Java

ASSIGNMENT No : 15

TITLE : Write a Java program to demonstrate exception handling using finally, throw, and throws.

Problem Statement

Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives

To implement finally, throw, and throws for effective exception management.

Theory

The finally block always executes regardless of whether an exception occurs, making it suitable for resource cleanup. The throw keyword explicitly generates an exception, while throws declares exceptions that a method may pass to its caller. These mechanisms improve structured error handling and application stability.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp15/Main1.java
Caught Exception : Age must be 18 or above
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp15/Main2.java
Exception in thread "main" java.lang.RuntimeException: Error : Division by zero occurred
    at Exp15.Main2.main(Main2.java:14)
Caused by: java.lang.ArithmeticException: / by zero
    at Exp15.Main2.main(Main2.java:10)
    at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:104)
    at java.base/java.lang.reflect.Method.invoke(Method.java:565)
    at jdk.compiler/com.sun.tools.javac.launcher.SourceLauncher.execute(SourceLauncher.java:258)
    at jdk.compiler/com.sun.tools.javac.launcher.SourceLauncher.run(SourceLauncher.java:138)
    at jdk.compiler/com.sun.tools.javac.launcher.SourceLauncher.main(SourceLauncher.java:76)
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Create a Login program that throws an exception for invalid password and uses finally block.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp15/Exercise/Exp1.java
Enter Password : 1234
Error: Invalid Password!
Login process completed.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp15/Exercise/Exp1.java
Enter Password : admin123
Login Successful.
Login process completed.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp15/Exercise/Exp2.java
Enter ATM PIN : 123
Error: Incorrect ATM PIN!
ATM Verification Completed!
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp15/Exercise/Exp2.java
Enter ATM PIN : 12345
Logging in...
ATM Check in Verified!!!
ATM Verification Completed!
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

This experiment helped in understanding the use of exception handling using `finally`, `throw`, and `throws` in Java. It demonstrated how exceptions can be generated, declared, and handled effectively to improve program reliability and prevent unexpected program termination. The use of the `finally` block ensured that important cleanup operations were executed regardless of whether an exception occurred. These concepts enhance error management, improve code readability, and are widely used in banking systems, login authentication, file handling, database applications, and other real-world Java applications where robust and secure exception handling is essential.